

The Independent-Row $\mathbf{K}_{(2,2)}$ Molecular Species Product

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Abstract

For a partition $\lambda \vdash 4$, let G_λ be the product of the cyclic groups generated by the disjoint standard cycles belonging to the parts of λ , and set $\mathbf{K}_\lambda = X^4/G_\lambda$. Thus the repeated parts in $(2, 2)$ rotate independently and $G_{(2,2)} = \langle (12), (34) \rangle \cong C_2^2$. We prove the natural species isomorphism

$$\mathbf{K}_{(2,2)} \times \mathbf{K}_{(2,2)} \cong 2\mathbf{K}_{(2,2)} \sqcup \mathbf{K}_{(1,1,1,1)}.$$

Consequently the coefficient vector, in partition order, is $(0, 0, 2, 0, 1)$. The expansion is unique both at the species level and after passing to cycle indices. The proof identifies all three relevant double cosets in $\mathfrak{S}_4$; an exact power-sum calculation provides a separate check.

1 Convention and result

Baolahy and Randrianirina record the molecular-species double-coset decomposition for products of the form X^n/H in [2, Proposition 2.2]. Their later preprint distinguishes a companion independent-row family $\mathbf{K}_\alpha = X^n/G_\alpha$ and asks for its product coefficients; it notes that the full subgroup lattice replaces the divisor lattice used in the cyclic case [1, Remark 3.2 and Open Problem 5.1].

We use that independent-row convention explicitly. If $\lambda = (\lambda_1, \dots, \lambda_k) \vdash 4$, let c_i be the standard cycle of length λ_i on the corresponding consecutive block and put

$$G_\lambda = \langle c_1 \rangle \times \dots \times \langle c_k \rangle, \quad \mathbf{K}_\lambda = X^4/G_\lambda,$$

where a one-cycle contributes the trivial group. This convention matters when parts repeat: here

$$H := G_{(2,2)} = \langle (12), (34) \rangle = \{e, (12), (34), (12)(34)\} \cong C_2^2.$$

It differs, for repeated equal parts, from the related convention in [2]. No assertion about a general formula for arbitrary α, β is made here.

Theorem 1. *Under the convention above,*

$$\mathbf{K}_{(2,2)} \times \mathbf{K}_{(2,2)} \cong \mathbf{K}_{(2,2)} \sqcup \mathbf{K}_{(1,1,1,1)} \sqcup \mathbf{K}_{(2,2)}.$$

Hence, in the order

$$(4), (3, 1), (2, 2), (2, 1, 1), (1, 1, 1, 1),$$

the integer coefficients are $(0, 0, 2, 0, 1)$. The five displayed molecular species have no nonzero integral linear relation. Their five cycle indices also have no nonzero integral linear relation.

2 Complete double-coset proof

Put $G = \mathfrak{S}_4$. Evaluation on a four-element set identifies X^4/H with the transitive G -set G/H . The diagonal action has the standard orbit decomposition

$$G/H \times G/H \cong \bigsqcup_{HgH \in H \backslash G/H} G/(H \cap gHg^{-1}). \quad (1)$$

We now determine every double coset in (1) without enumeration.

The only transpositions in H are (12) and (34). A permutation normalizing H must therefore preserve their unordered pair of supports, $\{\{1, 2\}, \{3, 4\}\}$. Conversely, every permutation preserving this two-block system normalizes H . Thus

$$N_G(H) = (\mathfrak{S}_{\{1,2\}} \times \mathfrak{S}_{\{3,4\}}) \rtimes \mathfrak{S}_2, \quad |N_G(H)| = 8.$$

The subgroup H is normal of index two in its normalizer. With $q = (13)(24)$, the normalizer is consequently the disjoint union of the two double cosets H and HqH . Each has four elements, and its intersection stabilizer in (1) is H .

Take $r = (23)$. Then

$$rHr^{-1} = \langle (13), (24) \rangle, \quad H \cap rHr^{-1} = 1.$$

The double-coset cardinality formula gives

$$|HrH| = \frac{|H|^2}{|H \cap rHr^{-1}|} = 16.$$

Moreover, HrH is disjoint from $N_G(H)$: if $h_1rh_2 \in N_G(H)$ for $h_1, h_2 \in H$, then $r \in N_G(H)$, whereas r does not preserve the two-block system. Since $|G \setminus N_G(H)| = 16$, it follows that

$$G = H \sqcup HrH \sqcup HqH.$$

These are therefore all the double cosets, with intersection stabilizers $H, 1, H$, respectively. Substitution into (1) yields

$$G/H \times G/H \cong G/H \sqcup G/1 \sqcup G/H,$$

and hence the asserted natural species isomorphism.

It remains to justify uniqueness. The Burnside group of a finite group is free abelian on its transitive G -sets: every finite G -set is the disjoint union of its orbits, and $G/L \cong G/L'$ exactly when L and L' are conjugate. In the present convention the five stabilizers are

λ	(4)	(3, 1)	(2, 2)	(2, 1, 1)	(1, 1, 1, 1)
G_λ	$\langle (1234) \rangle$	$\langle (123) \rangle$	$\langle (12), (34) \rangle$	$\langle (12) \rangle$	1.

Their orders distinguish every pair except the two groups of order four, and those are respectively cyclic and noncyclic. They are thus pairwise nonconjugate, proving species-level integral independence and uniqueness.

3 Cycle-index check

For $L \leq \mathfrak{S}_4$, the cycle index of X^4/L is

$$Z_L = \frac{1}{|L|} \sum_{\sigma \in L} p_{\lambda(\sigma)}.$$

With rows ordered as $p_4, p_3p_1, p_2^2, p_2p_1^2, p_1^4$ and columns ordered by the five partitions in Theorem 1, their coefficient matrix is

$$A = \begin{pmatrix} 1/2 & 0 & 0 & 0 & 0 \\ 0 & 2/3 & 0 & 0 & 0 \\ 1/4 & 0 & 1/4 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 & 0 \\ 1/4 & 1/3 & 1/4 & 1/2 & 1 \end{pmatrix}, \quad \det A = \frac{1}{24} \neq 0.$$

Thus these five cycle indices are linearly independent even over $\mathbb{Q}$, so in particular they have no nonzero integral relation.

For $H = G_{(2,2)}$ and the trivial subgroup, one has

$$Z_H = \frac{1}{4}(p_1^4 + 2p_2p_1^2 + p_2^2), \quad Z_1 = p_1^4.$$

Using $p_\lambda \star p_\mu = \delta_{\lambda\mu} z_\lambda p_\lambda$ and

$$z_{(1^4)} = 24, \quad z_{(2,1,1)} = 4, \quad z_{(2,2)} = 8,$$

gives the exact identity

$$Z_H \star Z_H = \frac{3}{2}p_1^4 + p_2p_1^2 + \frac{1}{2}p_2^2 = 2Z_H + Z_1.$$

This verifies the coefficients after applying cycle indices; the double-coset argument above proves the stronger species statement.

Computational and AI assistance disclosure

During preparation, a Python 3 program using exact integer and rational arithmetic enumerated all 24 elements of $\mathfrak{S}_4$ and checked the three double cosets, their intersection stabilizers, the associated table-of-marks solve, and the power-sum identity. The proof above does not rely on exhaustive enumeration. OpenAI Codex assisted with the finite search, proof checking, and manuscript preparation. The author reviewed the mathematical statements and assumes responsibility for the article.

References

- [1] Josaphat Baolahy and Benjamin Randrianirina. The molecular species $\mathbf{C}_\alpha$: Geometric realization and a closed formula for Kronecker coefficients, 2026. arXiv:2609.05010v1.
- [2] Josaphat W. V. Baolahy and Benjamin Randrianirina. Species, symmetric functions and Kronecker product. *Enumerative Combinatorics and Applications*, 6(3):Article S2R19, 2026.